

A harmonised point-extraction and comparison tool for heterogeneous paleoclimate datasets

Geoinformatics Project Report

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Contents

Abstract	1
1 Introduction	1
1.1 Why comparing paleoclimate models matters	1
1.2 Aim of this project	2
2 The Data	2
2.1 Trace21k	2
2.2 CHELSA-TraCE21k	3
2.3 Beyer 2021 (v4)	3
2.4 PalMod 2	4
3 Homogenization	5
3.1 Variables	5
3.2 Time	5
3.3 Space	5
4 Point Extraction	5
5 Results)	5
6 Future	6
7 Conclusion	6

Abstract

- paleoclimate simulations are very important for understanding earth systems - but they are distributed in diverse and sometimes incompatible formats
- different time origin and unit - spatial grid - conventions - temperature variables - time and spatial resolutions
- this project provides a reproducible python toolchain to download pre-process, and compare temperature archives for widely used datasets: - TraCE-21k, - CHELSA-TraCE21k-centennial, - Beyer2020/2021 - PalMod2 - harmonise to kiloyears-before-present (ka BP) scale - harmonise spatial grids
- convert temp units to degree celsius - extract temperature at user-defined point (use berlin as example)
- globally extract temperature of each dataset at each grid cell size of trace21k (most coarse), compare pair-wise
- all models agree on cold period around 20 kaBP followed by - but disagree in absolute temperature by several degrees.
- tried to find pattern in differences between dataset pair: global maps plotting the temp difference between each dataset-pair: - largest differences in high norther latitudes
- this report documents datasets, data acquisition method, preprocessing, harmonisation, data extraction at point and global comparison
- project is organized in several jupyter notebooks creating a full automatic pipeline Giersch 2026

1 Introduction

1.1 Why comparing paleoclimate models matters

- Paleoclimate models are useful for:
- Reconstructing Earth's climate before direct measurements existed
- Understanding how the climate responds to forcings like solar output, volcanic eruptions, orbital cycles, and CO₂
- Testing and validating modern climate models against known past states
- Establishing the natural variability range of climate, separate from human influence
- Constraining climate sensitivity (temperature change per CO₂ doubling)
- Studying planetary climate evolution — Mars, Venus, and other terrestrial bodies
- Using past warm periods (e.g. Pliocene) as analogues for near-future climate projections

- resolution time: decadal averages
- extend space: global land
- resolution space: T31_gx3
- source: Otto-Bliesner and Rosenbloom 2021

2.2 CHELSA-TraCE21k

- Chosen Version: CHELSA-TraCE21k-centennial
- developed by: see source
- year of development: see source
- how where they made: generated with the CHELSA-TraCE21k downscaling model.
- what was the goal:
- nr. files:
- size (gb):
- type: geotiff -> need to be converted to netcdf
- used variables: tasmin, tasmax -> $\text{tasmin} + \text{tasmax} / 2$ to get mean
- how to get: (via python script in my data_downloader.ipynb notebook)
- other info:
- extend time: Start Time 21k BP, End Time 0 BP
- resolution time: monthly averages, 100 year timesteps
- extend space: global
- resolution space: kilometer scale
- source: Karger et al. 2023, Karger et al. 2020, Karger et al. 2023

2.3 Beyer 2021 (v4)

- Chosen Version: 12293345 version 4 (figshare, published 24 June 2021)
- developed by: Robert M. Beyer, Mario Krapp & Andrea Manica (Department of Zoology, University of Cambridge)
- year of development: original paper published 14 July 2020 (received 03 July 2019); dataset v4 published 24 June 2021; addendum paper published 29 September 2021
- how were they made: Delta-method-based downscaling and bias correction, combining medium-resolution HadCM3 transient simulations with high-resolution HadAM3H snapshots and present-day observational data
- what was the goal: Fill the gap between high-temporal-resolution but low-spatial-resolution palaeoclimate simulations and high-spatial-resolution snapshot-only datasets

- nr. files:
- size (GB):
- type: NetCDF (data) + R / Python / MATLAB scripts
- used variables:
- how to get: (via python script in my data_downloader.ipynb notebook)
- other info: v4 adds global monthly NPP relative to v3; documented in the 2021 addendum
- extend time: 120000 BP to pre-industrial modern era (0 BP)
- resolution time: 2000-yr steps from 120000–22000 BP; 1000-yr steps from 22000 BP to pre-industrial (72 snapshots total)
- extend space: global terrestrial (full 360° longitude; latitude coverage 720x 300 cells at 0.5°, i.e. 150° of latitude)
- resolution space: 0.5° equirectangular grid (~55 km at the equator)
- source: R. M. Beyer, Krapp, and Manica 2020, R. M. Beyer, Krapp, and Manica 2021, R. Beyer 2020a, R. Beyer 2020b

2.4 PalMod 2

- Chosen Version: PalMod2 MPI-M MPI-ESM1-2-CR Transient Simulations of the Last Deglaciation with prescribed ice sheets from GLAC-1D reconstructions (r1i1p1f1)
- developed by: Mikolajewicz, U.; Kapsch, M.-L.; Gayler, V.; Meccia, V. L.; Riddick, T.; Ziemann, F. A.; Schannwell, C. (Max Planck Institute for Meteorology / DKRZ)
- year of development: dataset DOI published 2023
- how where they made: fully-coupled transient simulation with the Max Planck Institute Earth System Model version 1.2 in coarse resolution (MPI-ESM1-2-CR), combining the ECHAM6.3 spectral atmosphere (T31, 31 vertical levels) - what was the goal: produce a transient, fully-coupled simulation of the Last Deglaciation with realistic, time-evolving ice-sheet, topography and land-sea-mask forcing as part of the PalMod2 project
- nr. files:
- size (gb):
- type: NetCDF
- used variables: tas (PMMXMCRTDGr111Amtasgn30201, near-surface air temperature, monthly) tsl (PMMXMCRTDGr111Lmtslgn30201, soil temperature by layer, monthly)
- how to get: WDCC archive at DKRZ; download via jblob client after registration Climate and Klimarechenzentrum 2026 (via python script in my data_downloader.ipynb notebook)
- other info:

- extend time: 25000 BP to 0 BP (= 1950 CE)
- resolution time: variable-dependent — monthly for the standard atmosphere/land/ocean/sea-ice tables (Amon, Lmon, Omon, SImon, LImon, Emon), decadal (dec, Odec) and fixed (fx, Ofx); GLAC-1D ice-sheet/topography forcing updated every 10 years
- extend space: global
- resolution space: atmosphere/land T31 ($\sim 3,75^\circ$), ocean MPIOM1.6 nominal $\sim 3^\circ$
- source: Mikolajewicz et al. 2023

3 Homogenization

3.1 Variables

- which do i choose, are they directly comparable?

3.2 Time

- time extend
- time resolution

3.3 Space

- grid sizes
- changes in north south
- different resolutions for different times

4 Point Extraction

- do i just take the grid cell the point is in?

5 Results)

- For one example Point
- differences between models

6 Future

- Auto iterate over points and compare
- correlate with eg. north south, climate zones, water vs land etc.

7 Conclusion

- hard to compare due to differences eg. resolutions
- comparison can help to find out where they differ the most -> conclusion to how to improve models?

References

- Beyer, Robert (May 2020a). *Late Quaternary climate, bioclimate and vegetation*. DOI: 10.6084/m9.figshare.12293345.v4. URL: https://figshare.com/articles/dataset/LateQuaternary_Environment_nc/12293345.
- (May 2020b). *Late Quaternary climate, bioclimate and vegetation*. DOI: 10.6084/m9.figshare.12293345.v3. URL: https://figshare.com/articles/dataset/LateQuaternary_Environment_nc/12293345.
- Beyer, Robert M., Mario Krapp, and Andrea Manica (2021). “Addendum: High-resolution terrestrial climate, bioclimate and vegetation for the last 120,000 years”. In: DOI: 10.1038/s41597-021-01051-1. URL: <https://doi.org/10.1038/s41597-021-01051-1>.
- (2020). “High-resolution terrestrial climate, bioclimate and vegetation for the last 120,000 years”. In: DOI: 10.1038/s41597-020-0552-1. URL: <https://doi.org/10.1038/s41597-020-0552-1>.
- Climate, World Data Center for and Deutsches Klimarechenzentrum (2026). *Jblob Documentation*. <https://www.wdc-climate.de/ui/info?site=jblob>. Version 4.1. (Visited on 06/05/2026).
- Giersch, Levin Damian Antonin (2026). *Paleoclimate Model Comparison*. https://github.com/LevinGiersch/paleoclimate_model_comparison. Accessed: 2026-06-05.
- Karger, Dirk Nikolaus et al. (2023). “CHELSA-TraCE21k - high-resolution (1 km) down-scaled transient temperature and precipitation data since the Last Glacial Maximum”. In: DOI: 10.5194/cp-19-439-2023. URL: <https://doi.org/10.5194/cp-19-439-2023> (visited on 06/05/2026).
- (2020). *CHELSA-TraCE21k: Downscaled transient temperature and precipitation data since the last glacial maximum*. DOI: <http://dx.doi.org/10.16904/envidat.211>. URL: https://www.envidat.ch/dataset/chelsa_trace.
- Mikolajewicz, Uwe et al. (2023). *PalMod2 MPI-M MPI-ESM1-2-CR Transient Simulations of the Last Deglaciation with prescribed ice sheets from GLAC-ID reconstructions (r1i1p1f1)*. DOI: 10.26050/WDCC/PMMXMCRTDGP111. URL: <https://doi.org/10.26050/WDCC/PMMXMCRTDGP111>.
- Otto-Bliesner, Bette L. and Nan A. Rosenbloom (2021). *Simulation of the Transient Climate of the Last 21,000 Years (TraCE-21ka)*. DOI: 10.5065/CXB5-TV56. URL: <https://gdex.ucar.edu/datasets/d651050/> (visited on 06/05/2026).